Annex G. Glossary

Term	Definition
ADR	Architecture decision record: a short document explaining a design decision and its consequences
API	Interface that enables communication between applications
API Gateway	Single entry point that routes requests, validates the JWT and limits traffic
BFF	Backend-for-Frontend: a server layer tailored to the needs of one client
BLE	Bluetooth Low Energy; the mobile app uses it to provision the ESP32
DHT11	Digital sensor for air temperature and humidity
ESP32	Microcontroller with Wi-Fi and Bluetooth
Hexagonal (architecture)	Style that separates the domain from inbound and outbound adapters through ports
IoT	Internet of Things: connected physical devices that send data
JWT	Signed token that authenticates a user's requests
JWKS	Set of public keys that services use to verify JWT signatures
KY-038	Sound module with analog output for a relative noise level
Liquibase	Tool for versioning database changes
MQ-135	Gas and air-quality sensor; requires calibration and does not measure CO_2 in ppm without it
MQTT	IoT messaging protocol considered in the architecture; not confirmed in the firmware
OAuth2 / OIDC	Authorization and identity standards used for sign-in
OpenAPI	Standard for describing REST API contracts
Outbox	Pattern that stores events in the database before publishing them so they are not lost
RBAC	Role-based access control (SUPER_ADMIN, ADMIN, USER, VIEWER)
Smoke test	Quick end-to-end test confirming that the basic system responds
Testcontainers	Library that starts Docker containers (for example PostgreSQL) during tests
X-API-Key	Key an IoT device uses to authenticate when sending measurements